Reg. No. | | | | | | | | | | |

[12EC54/13EC54]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER –2015

EMBEDDED SYSTEM DESIGSN

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Explain the process of converting on assembly language into machine codes and obtaining the ROM image.	**08**
	b.	Explain how embedded systems are classified.	**06**
2.	**a.**	Explain the different target systems.	**04**
	b.	List out the features of locator.	**05**
	c.	Explain the need for appropriate OS- Hardware interface functions and also mention the platform dependency issues.	**05**
3.	**a.**	Explain I/O ports of PIC18F microcontroller.	**06**
	b.	Explain "STATUS" register in detail.	**08**
4.	**a.**	Explain the advantages of C language over assembly language.	**06**
	b.	Explain the following loop control statements with example. **i.** For statement. **ii.** While statement.	**08**
5.	**a.**	Write a C program for interfacing 16×2 LCD module with PIC microcontroller.	**07**
	b.	Explain watch dog timer and real time clock in detail.	**07**
6.	**a.**	With the help of neat block diagram. Explain serial peripheral interface (SPI).	**07**
	b.	Explain Inter integrated circuit bus with block diagram.	**07**
7.		Explain the following with necessary diagrams. **i.** Zigbee. **ii.** Universal serial bus. **iii.** RS232.	**14**
8.	**a.**	List out the design metrics used in the embedded system.	**06**
	b.	Differentiate between RISC and CISC architecture.	**08**

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